Part 4 of 6

OPTRONIC SENSORS: Gyroscopic Sensors

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Operational basics, salient features and military specific applications of ring laser gyroscopes and fibre-optic gyroscopes are discussed in this part of the article

Gyroscopic sensors constitute an essential component of inertial navigation sensors (INSes), which are extensively used on a range of aerospace vehicles. These include commercial airliners, military aircraft and spacecraft.

Another military application of optronic gyroscopic sensors is their use for mid-course correction of guided missiles. In the case of inertial navigation guidance, the vehicles use onboard sensors to determine their motion and acceleration with the help of gyroscopes and accelerometers. A gyroscope is used to measure angular rotation, whereas an accelerometer is used to measure linear motion. The two are combined into a single unit along with a control mechanism, and the unit is called inertial measurement unit (IMU), or INS.

Two very important laser-based gyroscopic sensors include ring laser gyroscope (RLG) and fibre-optic gyroscope (FOG). Operational basics, salient features and military-specific applications of the two types of optronic gyroscopic sensors are discussed in this part.

Ring laser gyroscope

Ring laser gyroscope, or RLG, is an important optronic rate sensor and, in essence, the heart of any INS. There are different types of gyroscopic sensors such as spinning wheel mechanical, dynamically-tuned, ring laser and fibre-optic.

RLG offers superior performance as compared to conventional spinning-type mechanical gyroscope. RLG has no moving parts and therefore no inherent drift terms due to absence of friction. Unlike a mechanical gyroscope, RLG does not resist changes to its orientation. A combination of three orthogonally-placed RLGs makes a rate sensor with three degrees of freedom.

RLG is primarily a rate sensor. It can be used to measure rotation by integrating rate information. It essentially consists of two counter-propagating laser beams over the same path. It operates on the principle of Sagnac effect, according to which null points of the internal standing-wave pattern produced by the counter-propagating laser modes shift in response to angular rotation. Shift in the nulls of the standing-wave pattern manifests itself in the form of a moving interference pattern observed by combining two laser modes.

The basic concept of RLG can be explained as follows. Frequency of oscillation in a linear laser is such that the laser cavity consists of an integral number of wavelengths. Linear laser cavity is constituted by double pass of the distance between two mirrors. Since it is imperative that the beam replicates itself for successive passes over the cavity length, there will always be nodes at the two mirrors, and the two oppositely-propagating laser beams comprise a standing wave.

In the case of a ring laser, two oppositely-directed laser beams can be considered as travelling waves, and there is no such constraint of having a node at the mirror. The two beams in this case can be independent of each other and oscillate at different amplitude and frequency. Oscillation frequency of each is determined by the optical path length and not the geometrical path length. Thus, any mechanism, rotation in this case, which produces an optical path difference, results in different frequencies of oscillation for the two laser beams.

If the rotation is in clockwise direction, the clockwise laser beam sees a larger optical path length than the path length encountered by the counter-clockwise beam. For rotation in counter-clockwise direction, it is the opposite. Frequency difference is proportional to the rotation rate of the cavity.

Thus, by measuring frequency difference, rotation rate of the cavity and the platform it is strapped on to can be measured. According to Sagnac equation, frequency difference is given by:

$$\Delta\nu = 4A\Omega / \lambda P = K\Omega$$

where A = encircled oriented surface area of the ring laser, P = perimeter, λ = wavelength and Ω = rotation rate. The parameter (K) is called scale factor, which is measured in Hz/radians/s.

Fig. 1 shows square ring laser geometry, which is one form of geometry in use. In the case of square ring laser geometry, four mirrors form a closed light path. A small part of the ring laser path houses the excited gain medium. Gain medium along with mirrors are responsible for producing two counter-propagating laser beams.